6 The electrical generator in a power station is driven by a steam turbine. The turbine absorbs thermal energy from a boiler and produces useful work. However, thermal energy must also be removed from the turbine by a cooling system as shown in Fig. 6.1.

The operating efficiency e of the turbine is defined by

$$e = \frac{\text{useful work output}}{\text{thermal energy input}}$$

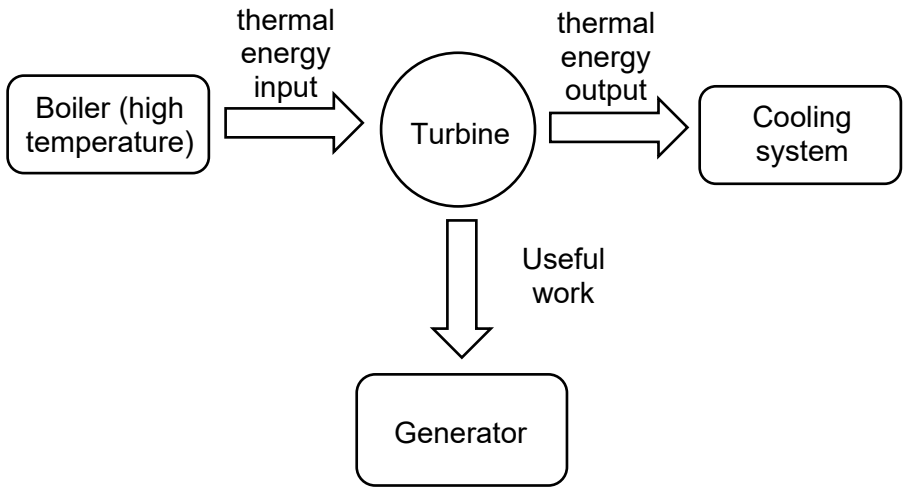

Fig. 6.1

The efficiency of heat engines, of which the turbine is an example, can never exceed a certain value which is fixed by temperatures of the boiler and the cooling system. This ideal efficiency e_{max} is given by the equation

$$e_{max} = \frac{T_2 - T_1}{T_2}$$

where T_2 is the thermodynamic temperature of the boiler and T_1 is the thermodynamic temperature of the cooling system.

Further data for a particular power station situated in Newtown, United Kingdom, are given in Fig. 6.2 below.

Electrical power output	200 MW
Efficiency of electrical generator	100%
Operating efficiency of turbine	31%
Ideal efficiency of turbine	52%
Effective temperature of cooling system (The cooling system uses water which enters at a temperature of 283 K and leaves at 291 K.)	330 K
Specific heat capacity of water in cooling system	4200 J kg^{-1} K^{-1}

Fig. 6.2

(a) Calculate the ideal efficiency e_{max} if the boiler temperature is 100 °C and the cooling system is at 27 °C.

e_{max} = ... [2]

(b) **(i)** Fig. 6.3 shows values of T_2 and e_{max} for a particular value of T_1.

T_2 / K	e_{max}
333	0.048
373	0.15
450	0.30
600	0.47
750	0.58

Fig. 6.3

Plot the variation with T_2 of e_{max} on the axes in Fig. 6.4.

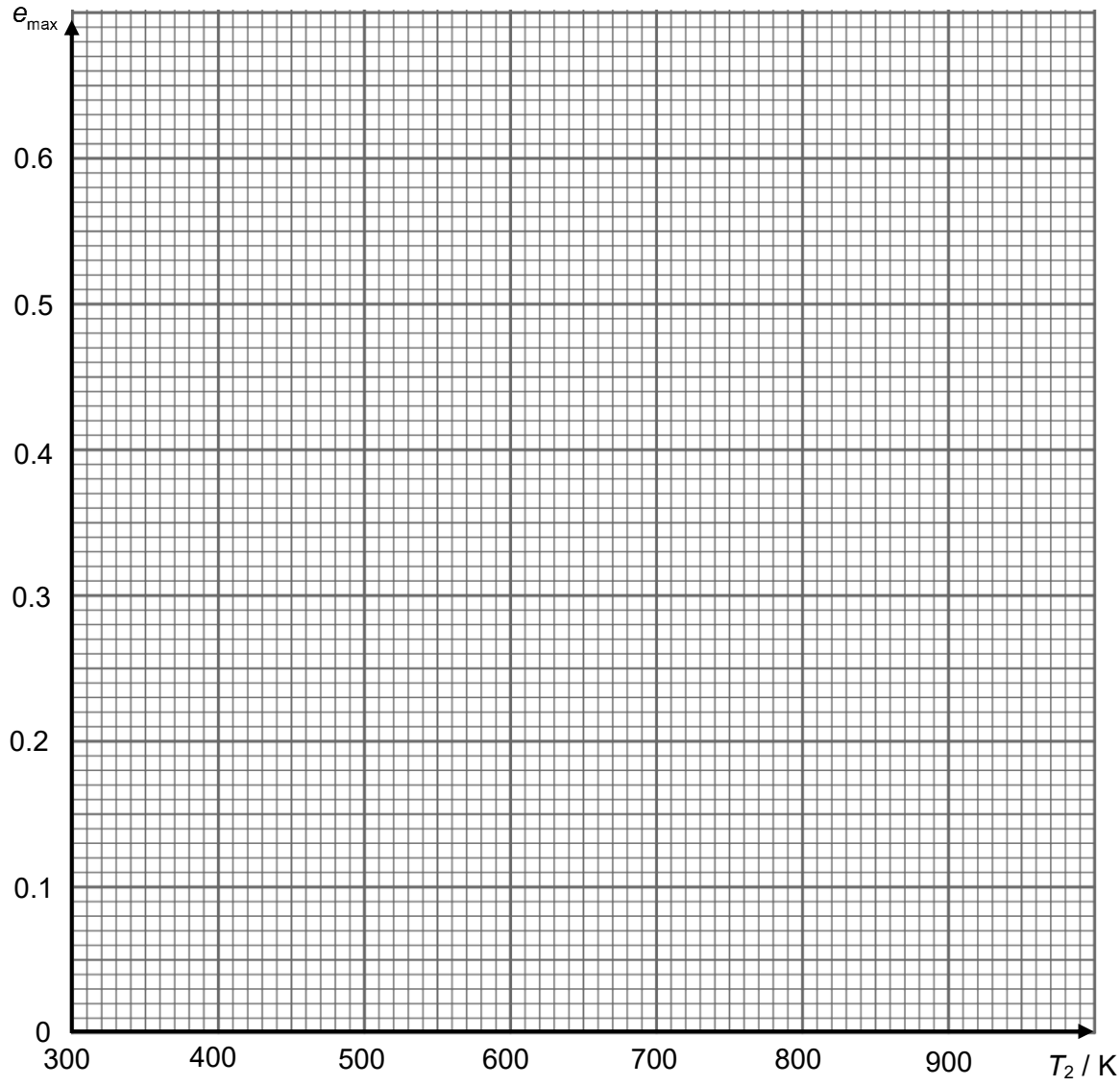

Fig. 6.4

[3]

(ii) From the graph, deduce the value of T_2 for which $e_{max} = 0$.

T_2 = ... K [1]

(iii) Hence, deduce the value of T_1.

T_1 = ... K [1]

(iv) Explain why it is not practical to attain an efficiency of 1.

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.....

..... [2]

(c) For the power station in Newtown, calculate

(i) the effective boiler temperature,

effective boiler temperature = ... K [1]

(ii) the rate of input of thermal energy to the turbine,

rate of input of thermal energy = ... W [1]

(iii) the rate at which thermal energy is removed from the turbine,

rate at which thermal energy is removed = ... W [1]

(iv) the required rate of flow of water through the cooling system.

rate of flow of water = ... kg s^{-1} [2]

(d) Suggest a reason for the discrepancy between the ideal efficiency of the turbine and its operating efficiency.

.....

..... [1]

(e) A significant fraction of the electrical power produced in the UK by burning fossil fuels is used for domestic heating. Two suggestions for improvement are as follows:

(i) Burn the fossil fuel in the home instead of at the power station.

(ii) Cogeneration or combined heat and power (CHP) mechanisms which use the thermal energy output from the turbine for domestic heating.

Comment critically on these suggestions.

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..... [2]

(f) Approximately 95% of Singapore's electricity is produced from fossil fuels (natural gas, coal, petroleum). Due to the lack of natural resources, most of the fossil fuels are imported from neighbouring countries. To enhance the nation's energy security and reduce our carbon footprint, the government has been exploring alternative energy sources.

(i) Suggest, with a reason, the renewable energy source that is the most viable option for Singapore.

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..... [2]

(ii) State and explain a limitation in the deployment of the renewable energy source suggested in **(f)(i)** on a large scale to generate electricity reliably in Singapore.

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..... [1]

End of Paper 2